

UNIVERSITY OF MIAMI
Curriculum Vitae

1. Date: 14 September 2026

PERSONAL

2. Name: Michael S. Fischer
3. Email Address: mike.fischer@miami.edu
4. Current Academic Rank: Assistant Professor
5. Primary Department: Atmospheric Sciences
6. Secondary or Joint Appointments: Frost Institute for Data Science and Computing (IDSC)

HIGHER EDUCATION

7. Institutional (institution; degree; date conferred):

University at Albany – State University of New York; Ph.D., Atmospheric Science;
June 2018

Dissertation: Tropical Cyclone Rapid Intensification in Environments of Upper-Tropospheric Troughs: Environmental Influences and Convective Characteristics.
Advisors: Drs. Brian Tang and Kristen Corbosiero

Florida International University; B.S., Geosciences; 2013
8. Non-Institutional (description; dates): N/A
9. Certification, licensure (description; board or agency; dates): N/A

EXPERIENCE

10. Academic (institutions; rank/status; dates):

University of Miami; Assistant Professor; August 2024–Present

University at Albany – State University of New York; Teaching Assistant; 2013–2016
11. Non-Academic (employers; title; responsibilities; dates):

University of Miami’s Cooperative Institute for Marine and Atmospheric Studies (CIMAS); Associate Scientist; Responsibilities included writing proposals, writing peer-reviewed publications, leading and co-leading research projects; 2023–2024

University of Miami's Cooperative Institute for Marine and Atmospheric Studies (CIMAS); Assistant Scientist; Responsibilities included writing proposals, writing peer-reviewed publications, leading and co-leading research projects; 2020–2023

NOAA's Atlantic Oceanographic and Meteorological Laboratory, Hurricane Research Division; National Research Council (NRC) Postdoctoral Research Associate; Performed research on tropical cyclone structure and intensity change using aircraft reconnaissance observations; 2018–2020

Innovim / NOAA National Hurricane Center; Scientific Programmer; 2018

12. Military (branch; rank; responsibilities; dates): N/A

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PUBLICATIONS [author(s) (in actual precedence of authorship); title; publisher or journal name; date (current year first); page numbers]

13. Books and monographs published: N/A

14. Juried or refereed journal articles and exhibitions:

Fischer, M. S., G. R. Alvey III, D. Jariwala, and P. D. Reasor, 2026: To align or not to align? That is the question. *J. Geophys. Res. Atmos.*, **131**, e2025JD045986, <https://doi.org/10.1029/2025JD045986>.

Nolan, D. S., M. S. Fischer, and M. E. O'Neill, 2025: Reconsideration of the mass and condensate sources for the tropical cyclone outflow. *Bull. Amer. Meteor. Soc.*, **106**, E1342–E1359.

Fischer, M. S., P. D. Reasor, J. P. Dunion, and R. F. Rogers, 2025: Are rapidly intensifying tropical cyclones associated with unique vortex and convective characteristics? *Mon. Wea. Rev.*, **153**, 183–203.

Rios-Berrios, R., P. M. Finocchio, J. J. Alland, X. Chen, M. S. Fischer, S. Stevenson, and D. Tao, 2024: A review of the effects of vertical wind shear on tropical cyclone structure and intensity. *J. Atmos. Sci.*, **81**, 713–741.

Shimada, U., P. Reasor, R. Rogers, M. S. Fischer, F. Marks, J. Zawislak, and J. Zhang, 2024: Shear-relative asymmetric kinematic characteristics of intensifying hurricanes as observed by airborne Doppler radar. *Mon. Wea. Rev.*, **152**, 491–512.

Fischer, M. S., R. F. Rogers, P. D. Reasor, and J. P. Dunion, 2024: An observational analysis of the relationship between tropical cyclone vortex tilt, precipitation structure, and intensity change. *Mon. Wea. Rev.*, **152**, 203–225.

Wadler, J. B., J. J. Cione, R. F. Rogers, and M. S. Fischer, 2023: On the distribution of convective and stratiform precipitation in tropical cyclones from airborne Doppler radar and its relationship to intensity change and environmental wind shear direction. *Mon. Wea. Rev.*, **151**, 3209–3233.

Wadler, J. B., J. E. Rudzin, B. J. de la Cruz, J. Chen, M. S. Fischer, G. Chen, N. Qin, B. Tang, and Q. Li, 2023: A review of recent research progress on the external influences of tropical cyclone intensity change. *Tropical Cyclone Res. and Rev.*, <https://doi.org/10.1016/j.tcr.2023.09.001>.

DesRosiers, A. J., M. M. Bell, P. J. Klotzbach, M. S. Fischer, and P. Reasor, 2023: Observed relationships between tropical cyclone vortex height, intensity, and intensification rate. *Geophys. Res. Lett.*, **50**, e2022GL101877.

Stone, Z., G. R. Alvey III, J. P. Dunion, M. S. Fischer, D. J. Raymond, R. F. Rogers, S. Sentic, and J. Zawislak, 2023: Thermodynamic contribution to vortex alignment and rapid intensification of Hurricane Sally (2020). *Mon. Wea. Rev.*, **151**, 931–951.

Fischer, M. S., P. D. Reasor, B. H. Tang, K. L. Corbosiero, R. D. Torn, and X. Chen, 2023: A tale of two vortex evolutions: Using a high-resolution ensemble to assess the impacts of ventilation on a tropical cyclone rapid intensification event. *Mon. Wea. Rev.*, **151**, 297–320.

Hazelton, A., G. J. Alaka, M. S. Fischer, R. D. Torn, and S. Gopalakrishnan, 2023: Factors influencing the track of Hurricane Dorian (2019) in the West Atlantic: Analysis of a HAFS ensemble. *Mon. Wea. Rev.*, **151**, 175–192.

Fischer, M. S., P. D. Reasor, R. F. Rogers, and J. F. Gamache, 2022: An analysis of tropical cyclone vortex and convective characteristics in relation to storm intensity using a novel airborne Doppler radar database. *Mon. Wea. Rev.*, **150**, 2255–2278.

Alvey, G., M. S. Fischer, P. Reasor, J. Zawislak, and R. Rogers, 2022: Processes underlying the vortex repositioning during Dorian's (2019) early stages that increased its favorability for rapid intensification. *Mon. Wea. Rev.*, **150**, 193–213.

Zawislak, J., R. Rogers, S. Aberson, G. Alaka, G. Alvey, A. Aksoy, L. Bucci, J. Cione, N. Dorst, J. Dunion, M. S. Fischer, J. Gamache, S. Gopalakrishnan, A. Hazelton, H. Holbach, J. Kaplan, H. Leighton, F. Marks, S. Murillo, P. Reasor, K. Ryan, K. Sellwood, J. Sippel, and J. Zhang, 2022: Accomplishments of NOAA's Airborne Hurricane Field Program and a broader future approach to forecast improvement. *Bull. Amer. Meteor. Soc.*, **103**, E311–E338.

Hazelton, A., G. J. Alaka, L. Cowan, M. S. Fischer, and S. Gopalakrishnan, 2021: Understanding the processes causing the early intensification of Hurricane Dorian through an ensemble of the Hurricane Analysis and Forecast Systems (HAFS). *Atmosphere*, **12**, 93.

Fischer, M. S., R. F. Rogers, and P. D. Reasor, 2020: The rapid intensification and eyewall replacement cycles of Hurricane Irma (2017). *Mon. Wea. Rev.*, **148**, 981–1004.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero, 2019: A climatological analysis of tropical cyclone rapid intensification in environments of upper-tropospheric troughs. *Mon. Wea. Rev.*, **147**, 3693–3719.

Fischer, M. S., B. H. Tang, K. L. Corbosiero, and C. M. Rozoff, 2018: Normalized convective characteristics of tropical cyclone rapid intensification events in the North Atlantic and eastern North Pacific basins. *Mon. Wea. Rev.*, **146**, 1133–1155.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero, 2017: Assessing the influence of upper-tropospheric troughs on tropical cyclone intensification rates after genesis. *Mon. Wea. Rev.*, **145**, 1295–1313.

15. Other works, publications and abstracts:

Book/Encyclopedia Chapters

Sippel, J. A., and M. S. Fischer, 2026: Tropical cyclone intensity predictability. Reference Module in Earth Systems and Environmental Sciences, *Encyclopedia of*

Atmospheric Sciences, Third Edition, Vol. 1, 446–459,
<https://doi.org/10.1016/B978-0-323-96026-7.00140-5>.

Manuscripts in Preparation

Alvey III, G. R., J. Zhang, J. Dunion, B. Dahl, and M. S. Fischer, 2026: The path to alignment: Dual-Doppler and SAMURAI observations of Tropical Cyclone Fiona (2022). *Mon. Wea. Rev.*, in preparation.

Fischer, M. S., W. Bouza, D. Jariwala, and M. Reardon, 2026: Estimating tropical cyclone vortex tilt from geostationary satellite imagery using a random forest. *Wea. Forecasting*, in preparation.

16. Other works accepted for publication: N/A

PROFESSIONAL

17. Funded Research Performed (include all grants received in the last five years, identifying the principal investigator and the amounts and dates of the awards):

Collaborative Research: Tropical Cyclones from 400 to 40 hPa. National Science Foundation, 2/2024–1/2027, \$777,109. (Co-PI with Drs. David Nolan and Morgan O’Neill)

Application of Machine Learning Techniques to the Automated Quality Control of Airborne Doppler Radar Data Used in Operational Analysis of Hurricane Winds and Reflectivity. NOAA-OAR-WPO-2023-2007516, 8/2023–7/2025, \$400,686. (PI with Co-PI Dr. Michael Bell)

A comprehensive observational examination of the physical processes that link tropical cyclone vortex alignment to future intensity change. National Science Foundation, 4/2023–3/2026, \$539,622. (Co-PI with Drs. George Alvey and David Nolan)

18. Editorial responsibilities:

Associate Editor for *Monthly Weather Review* (2020–Present)

19. Professional and Honorary Organizations (member; officer; date):

- Member of the American Meteorological Society (2014–Present)
- Member of the American Geophysical Union (2025–Present)

20. Honors and Awards:

- Advances in Atmospheric Sciences Editor’s Award (2026)
- Journal of the Atmospheric Sciences Editor’s Award (2024)
- Quarterly Journal Prize Reviewer’s Certificate (2020)
- University at Albany Distinguished Dissertation Award (2019)
- Narayan R. Gokhale Distinguished Research Scholarship Award (2018)

- Outstanding Student Oral Presentation Award, *33rd Conference on Hurricanes and Tropical Meteorology* (2018)
- Top graduate student forecaster for Green Bay, WI in the national WxChallenge competition (2015)

21. Post-Doctoral Fellowships:

National Research Council Postdoctoral Research Associate at NOAA's Atlantic Oceanographic and Meteorological Laboratory (2018–2020)

22. Other Professional Activities (e.g., papers presented; performances; conference proceedings; seminar or conference panel member; catalogue work; etc.):

Fischer, M. S., W. Bouza, D. Jariwala, and M. Reardon: A machine learning approach to estimate the three-dimensional tropical cyclone wind field. 37th Conference on Hurricanes and Tropical Meteorology, San Diego, CA, April 2026.

Fischer, M. S.: A machine learning approach to estimate the three-dimensional tropical cyclone kinematic structure (invited presentation). NSF IPAM Workshop on Mathematics and Machine Learning for Earth System Simulation, Los Angeles, CA, February 2026.

Fischer, M. S.: A machine learning approach to estimate the three-dimensional tropical cyclone wind field (invited presentation). 2025 American Geophysical Union Annual Meeting, New Orleans, LA, December 2025.

Fischer, M. S.: Exploring tropical cyclone structural precursors to rapid intensification (invited presentation). 2025 Symposium for Tropical Cyclone Risk in a Changing Climate, Clearwater, FL, June 2025.

Fischer, M. S.: A machine learning approach to estimate tropical cyclone structure using geostationary satellite observations. Working Group on Tropical Meteorology Research (WGTMR) Annual Meeting, Virtual, May 2025.

Fischer, M. S.: Using machine learning to better understand tropical cyclone structure and intensity change. Seminar for University of Miami's Frost Institute for Data Science and Computing, Miami, FL, April 2025.

Fischer, M. S.: Data-driven insights into tropical cyclone rapid intensification. Seminar for the University of Alabama in Huntsville, Huntsville, AL, March 2025.

Fischer, M. S., P. D. Reasor, J. P. Dunion, and R. F. Rogers: Are rapidly intensifying tropical cyclones associated with unique vortex characteristics? American Meteorological Society's 36th Conference on Hurricanes and Tropical Meteorology, Long Beach, CA, May 2024.

Fischer, M. S., R. F. Rogers, P. D. Reasor, and J. P. Dunion: The relationship between tropical cyclone vortex tilt, precipitation structure, and intensity change. American Meteorological Society's 104th Annual Meeting, Baltimore, MD, January 2024.

Fischer, M. S., P. D. Reasor, R. F. Rogers, J. P. Dunion, B. H. Tang, K. L. Corbosiero, R. D. Torn, and X. Chen: Exploring the relationship between vortex misalignment and tropical cyclone intensity change. Hurricane Forecast Improvement Project seminar, April 2023.

Fischer, M. S., P. D. Reasor, R. F. Rogers, and J. F. Gamache: An analysis of tropical cyclone vortex and convective characteristics in relation to storm intensity using a novel airborne Doppler radar database. 35th Conference on Hurricanes and Tropical Meteorology, New Orleans, LA, May 2022.

Fischer, M. S., R. F. Rogers, P. D. Reasor, and J. P. Dunion: Relationships between vortex tilt, convective structure, and intensity change in early-stage tropical cyclones. 34th Conference on Hurricanes and Tropical Meteorology, May 2021.

Fischer, M. S., R. F. Rogers, P. D. Reasor, and J. P. Dunion: How is tropical cyclone vortex tilt related to precipitation structure and intensity change? 14th International Conference on Mesoscale Convective Systems and High-Impact Weather in East Asia, April 2021.

Fischer, M. S., R. F. Rogers, and P. D. Reasor: An examination of local shear, vortex tilt, and tropical cyclone intensity change using airborne radar observations. American Meteorological Society's 100th Annual Meeting, Boston, MA, January 2020.

Fischer, M. S., R. F. Rogers, and P. D. Reasor: The rapid intensification and eyewall replacement cycles of Hurricane Irma (2017). 19th Cyclone Workshop, Seeon, Germany, October 2019.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero: Characteristics of tropical cyclone rapid intensification in environments of upper-tropospheric troughs. 33rd Conference on Hurricanes and Tropical Meteorology, Ponte Vedra Beach, FL, April 2018.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero: Convective characteristics of tropical cyclone rapid intensification in environments of upper-tropospheric troughs. 8th Northeast Tropical Meteorology Workshop, Rensselaerville, NY, June 2017.

Fischer, M. S., B. H. Tang, and K. L. Corbosiero: The influence of an upper-tropospheric potential vorticity anomaly on rapid tropical cyclogenesis. 32nd Conference on Hurricanes and Tropical Meteorology, San Juan, Puerto Rico, April 2016.

TEACHING

23. Teaching Awards Received: N/A

24. Teaching Specialization (Note briefly courses taught, new courses developed, innovative or experimental teaching etc.):

- RSM 780 – Directed Readings in Tropical Weather and Climate (Instructor; Spring 2025)

- RSM 780 – Machine Learning Applications in the Atmospheric Sciences (Instructor; Fall 2026)
 - ATM 102 – Introduction to Weather and Climate (Instructor; Fall 2025)
 - ATM 411 – Research in Atmospheric Science (Thesis/Individual Study) (Instructor; Fall 2025–Fall 2026)
 - ATM 412 – Research in Atmospheric Science (Thesis/Individual Study) (Instructor; Spring 2026–Fall 2026)
 - ATM 303 – Meteorological Instrumentation and Observation (guest lecture; Spring 2022, 2025)
 - ATM 536 – Hurricanes (guest lecture; Spring 2024)
 - ATM 652 – Introduction to Atmospheric Physics (guest lecture; Fall 2024, 2025)
 - Weather Discussion I, St. Cloud State University (guest lecture)
 - Teaching Assistant, University at Albany – State University of New York (2013–2016)
25. Thesis and Dissertation Advising/Post-doctoral student supervision (chairman or committee member; topic; student name; date):
- Kelly Neighbour (Ph.D. student, University of Miami, 2025–Present) – Advisor
 - Helena Tsigos (Ph.D. student, University of Miami, 2025–Present) – Advisor
 - Cameron Pine (Ph.D. student, University of Miami, 2023–Present) – Co-advisor with Drs. David Nolan and George Alvey
 - Aidan Mahoney (Ph.D. student, University of Miami, Defended April 2026) – Committee member
 - Matthew Donahue (M.S. student, University of Miami, Defended May 2026) – Committee member
 - Shannon McCloskey (M.P.S., University of Miami, 2026) – Research advisor. Thesis: *An Observational Analysis of Rapidly Intensifying Tropical Cyclones Undergoing Eyewall Replacement Cycles*
 - Amanda Keane (M.P.S., University of Miami, 2022) – Research advisor. Thesis: *Using NOAA Tail Doppler Radar Observations to Examine the Evolution of Vortex Tilt in Rapidly Intensifying Hurricane Ida (2021)*
 - Deelan Jariwala (Undergraduate student, University of Miami, 2026) – Undergraduate thesis advisor
 - Matthew Reardon (Undergraduate student, University of Miami, 2026) – Undergraduate thesis advisor
 - Arielle Jorge (Undergraduate student, University of Miami, 2025–Present) – Undergraduate thesis advisor

SERVICE

26. University Committee and Administrative Responsibilities:

- 2027 Rosenstiel Award Committee Chair

27. Community Activities:

- Developer and data steward of the Tropical Cyclone Radar Archive of Doppler Analyses with Recentering (TC-RADAR) database, which contains over 1,600 airborne Doppler radar analyses from storms sampled by NOAA's WP-3D aircraft (2019–Present)
- Developer of TC-ATLAS, an open-source, interactive web platform for the visualization and analysis of tropical cyclone structure and intensity change: <https://tcatlas.org/> (2025–Present)
- NOAA Hurricane Research Division Field Program science team member (2018–Present); Principal Investigator of the Vortex Alignment Module in the annual hurricane field program
- Reviewer for *Journal of the Atmospheric Sciences*, *Monthly Weather Review*, *Journal of Applied Meteorology and Climatology*, *Quarterly Journal of the Royal Meteorological Society*, *Atmosphere*, *Journal of Geophysical Research – Atmospheres*, *Geophysical Research Letters*, *Dynamics of Atmospheres and Oceans*, *npj Climate and Atmospheric Science*, and *Weather and Climate Dynamics* (2017–Present)
- Proposal reviewer for the National Science Foundation and NASA (2023–Present)
- A career in hurricane science – South Miami Heights Elementary School outreach event (2024–Present)
- Rapporteur for the *11th International Workshop on Tropical Cyclones* (2026)
- Session chair for *Tropical Cyclone Rapid Intensification* at the *35th–37th Conference on Hurricanes and Tropical Meteorology* (2022–2026)
- Session co-convenor at the *37th Conference on Hurricanes and Tropical Meteorology* (2025–2026)
- NOAA Hurricane and Ocean Testbed (HOT) collaborator (2021–2024)
- Rapid Intensification in Tropical Cyclones (TCRI) science team member (2020–2024)
- Working group member for the *10th International Workshop on Tropical Cyclones* (2022)
- Max Eaton Award committee member at the *34th and 35th Conference on Hurricanes and Tropical Meteorology* (2021, 2022)
- Student poster award committee member at the *34th Conference on Hurricanes and Tropical Meteorology* (2022)
- Session co-chair for *Rapid Intensification of Tropical Cyclones* at the 101st American Meteorological Society Annual Meeting (2021)